



SIMATS Online Programming Lab

C PROGRAMMING



M Nishanth
192511154RB

CO3 - AT1 - Analytical Problem solving

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QUESTIONS

Q1

CO3 - AT1 - Analytical Problem Solving
20 marks

**Q2**

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20 marks

**Q3**

CO3 - AT1 - Analytical Problem Solving
20 marks

**Q4**

CO3 - AT1 - Analytical Problem Solving
20 marks

**Q5**

CO3 - AT1 - Analytical Problem Solving
20 marks



5/5 answered

Q1 CO3 - AT1 - Analytical Problem Solving 20 marks

1. Given an $N \times N$ matrix, Write a C Program to determine whether it is:

- a) Symmetric
- b) Upper triangular
- c) Lower triangular
- d) Diagonal matrix
- e) Identity matrix
- f) Display all applicable classifications.

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n, a[100][100];
5     int i, j;
6     int symmetric = 1, upper = 1, lower = 1;
7     int diagonal = 1, identity = 1;
8
9     scanf("%d", &n);
10
11     for (i = 0; i < n; i++) {
12         for (j = 0; j < n; j++) {
13             scanf("%d", &a[i][j]);
14         }
15     }
```

```
16
17
18     for (i = 0; i < n; i++) {
19         for (j = 0; j < n; j++) {
20
21
22             if (a[i][j] != a[j][i])
23                 symmetric = 0;
24
25
26             if (i > j && a[i][j] != 0)
27                 upper = 0;
28
29
30             if (i < j && a[i][j] != 0)
31                 lower = 0;
32
33
34             if (i != j && a[i][j] != 0)
35                 diagonal = 0;
36
37
38             if (i == j && a[i][j] != 1)
39                 identity = 0;
40
41             if (i != j && a[i][j] != 0)
42                 identity = 0;
43         }
44     }
45
46     if (symmetric)
47         printf("Symmetric Matrix\n");
48
49     if (upper)
50         printf("Upper Triangular Matrix\n");
51
52     if (lower)
53         printf("Lower Triangular Matrix\n");
54
55     if (diagonal)
56         printf("Diagonal Matrix\n");
57
58     if (identity)
59         printf("Identity Matrix\n");
60
```

```
61     return 0;  
62 }
```

Input

```
3  
1 0 0  
0 1 0  
0 0 1
```



Output

**Run****Save**